

The Wonderful World of Active Many-Particle Systems

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Abstract. Since the subject of traffic dynamics has captured the interest of physicists, many astonishing effects have been revealed and explained. Some of the questions now understood are the following: Why are vehicles sometimes stopped by so-called “phantom traffic jams”, although they all like to drive fast? What are the mechanisms behind stop-and-go traffic? Why are there several different kinds of congestion, and how are they related? Why do most traffic jams occur considerably before the road capacity is reached? Can a temporary reduction of the traffic volume cause a lasting traffic jam? Why do pedestrians moving in opposite directions normally organize in lanes, while nervous crowds are “freezing by heating”? Why do panicking pedestrians produce dangerous deadlocks? All these questions have been answered by applying and extending methods from statistical physics and non-linear dynamics to self-driven many-particle systems.

1 Introduction

Systems far from equilibrium are characterized by not being closed, but exchanging energy, particles, and information with their environment. Their dynamics is often very complex, and usually there are no general results like the ones from thermodynamics and statistical physics for gases, fluids, and solids in their stationary equilibrium state. This is partly due to the fact that non-equilibrium systems can be quite different in nature, dependent on the respective interactions with their environment.

Nevertheless, there exists a class of non-equilibrium systems for which there is some hope for a unified description, despite of the rich variety of observed phenomena. I am talking about self-driven or active many-particle systems, in which the particles are in some sense “motorized” [1,2,3,4]. Examples are animal herds, flocks of birds, or traffic. Sometimes, one can also apply this approach to socio-economic systems, where shares, goods, or individuals correspond to the particles. Surprisingly enough, the most important transport properties of such “particles” can be often taken into account by two simple generalizations of Newton’s equation of motion.

2 Beyond Newton

Newton's equation of motion from classical mechanics describes the acceleration $\ddot{\mathbf{x}}_i(t)$ of a body i of mass m_i subject to pair interactions with other bodies j :

$$m_i \ddot{\mathbf{x}}_i(t) = \sum_{j(\neq i)} \mathbf{f}_{ij}(t). \quad (1)$$

The interaction forces $\mathbf{f}_{ij}(t)$ are mostly dependent on the locations $\mathbf{x}_i(t)$ and $\mathbf{x}_j(t)$ of the interacting bodies i and j at time t . Often, they depend only on the distance vector $\mathbf{d}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)$, but in special cases, they are also functions of the velocities $\mathbf{v}_i(t) = \dot{\mathbf{x}}_i(t)$ and $\mathbf{v}_j(t) = \dot{\mathbf{x}}_j(t)$. For potential forces, the above many-body system can be characterized by a Hamiltonian. A typical example is the description of the motion of celestial bodies.

In *driven* many-particle systems such as fluids under the influence of pressure gradients and boundary forces, or vibrated granular media like sand, we need to consider additional interactions with the environment. This concerns (external) driving forces $\mathbf{f}_0(\mathbf{x}, t)$ due to boundary interactions and gravitational or electrical fields, dissipative friction forces $\mathbf{f}_i^{\text{fr}}(t) = -\gamma_i \mathbf{v}_i(t)$ with friction coefficient γ_i , and individual fluctuations $\boldsymbol{\zeta}_i(t)$ reflecting thermal interactions with the environment (boundaries, air, etc.) or a variation of the surface structure of the particles:

$$m_i \ddot{\mathbf{x}}_i(t) = \mathbf{f}_0(\mathbf{x}_i(t), t) - \gamma_i \mathbf{v}_i(t) + \sum_{j(\neq i)} \mathbf{f}_{ij}(t) + \boldsymbol{\zeta}_i(t). \quad (2)$$

In such systems, the ongoing competition between the driving forces and the dissipative friction forces leads to a spatio-temporal redistribution of energy, which produces a great variety of self-organization phenomena. This results from non-linearities in the equations of motion, which allow small initial perturbations to be enhanced and non-equilibrium patterns to be dynamically stabilized.

In the following, we will generalize Eq. (2) only a little. In contrast to conventional physical systems, we will assume to be confronted with “self-driven”, “active”, or “motorized” particles with an internal energy reservoir [5]. Therefore, the interaction forces $\mathbf{f}_{ij}(t)$ do not necessarily fulfill the 3rd Newtonian law $\mathbf{f}_{ji}(t) = -\mathbf{f}_{ij}(t)$ (*actio = reactio*). Additionally, each particle can display an individual driving force $\mathbf{f}_i^0(t)$ instead of $\mathbf{f}_0(\mathbf{x}_i, t)$. We will see that this will imply various interesting phenomena, which can be observed in living systems. In this connection, it is useful to rewrite Eq. (2) somewhat, where the accelerations $\mathbf{a}_{ij}(t)$ are often called “generalized forces”:

$$\dot{\mathbf{v}}_i(t) = \frac{v_i^0(t) \mathbf{e}_i^0(t) + \boldsymbol{\xi}_i(t) - \mathbf{v}_i(t)}{\tau_i} + \sum_{j(\neq i)} \mathbf{a}_{ij}(t). \quad (3)$$

Here, we have used the notations $\mathbf{f}_i^0(t) = \gamma_i v_i^0(t) \mathbf{e}_i^0$, $\gamma_i = m_i/\tau_i$, $\mathbf{f}_{ij}(t) = m_i \mathbf{a}_{ij}(t)$, and $\boldsymbol{\zeta}_i(t) = \gamma_i \boldsymbol{\xi}_i(t)$. Equation (3) can be also applied, when the masses m_i are not well-defined, as in socio-economic systems. It implies that, while subject to the fluctuations $\boldsymbol{\xi}_i(t)$, the velocity $\mathbf{v}_i(t)$ is adapted to the equilibrium velocity v_i^0 and the direction $\mathbf{e}_i^0$ of motion, if particle i is far enough away from other particles j (i.e., $\mathbf{a}_{ij} = 0$). The adaptation is exponential in time with the relaxation time τ_i . In the following, we will assume that the interaction forces are short-ranged and repulsive. Moreover, we will focus on the question, what kinds of different phenomena occur in self-driven many-particle systems under which conditions.

3 Vehicle Traffic on Freeways

Let us now face freeway traffic as an example for one-dimensional non-equilibrium systems, focussing on the longitudinal dynamics within a single lane. In that case, the directions $\mathbf{e}_i^0$ of motion of the vehicles (playing the role of the active particles) are all parallel. One frequently assumes that the vehicles i mainly react to the respective leading vehicles ($i - 1$), while the response to the following vehicle ($i + 1$) is neglected. In such car-following models, one applies equations of the form

$$\frac{dv_i(t)}{dt} = \frac{v_i^0 - v_i(t)}{\tau_i} + a_{i(i-1)}(t), \quad (4)$$

where we drop individual fluctuations for the moment. In simple models, the deceleration $a_{i(i-1)}(t) \leq 0$ depends only on the distance $s_i(t) = [x_{i-1}(t) - x_i(t)]$, but in reality, it is also a function of the own speed $v_i(t)$ and the relative velocity (approaching rate) $\Delta v_i(t) = [v_i(t) - v_{i-1}(t)]$ to the vehicle in front. There are, of course, also more detailed approaches, but these violate Newton's 3rd law *actio = reactio* as well. In the following, we will assume identical driver-vehicle units, so that we have $v_i^0 = v_0$, $\tau_i = \tau$, and $a_{i(i-1)}(s_i, v_i, \Delta v_i) = a(s_i, v_i, \Delta v_i)$. Then, it is often useful to introduce the traffic-dependent “optimal velocity”

$$V(s_i, v_i, \Delta v_i) = v_0 + \tau a(s_i, v_i, \Delta v_i) \leq v_0, \quad (5)$$

which corresponds to the stationary solution of Eq. (4) with $dv_i/dt = 0$. Driver i tries to adapt his actual velocity $v_i(t)$ to this velocity within the relaxation time τ .

3.1 “Phantom Traffic Jams”

If the “optimal velocity” depends only on the distance $s_i(t)$, we find the following fundamental relation between the traffic flow and the vehicle density $\rho = 1/s$ in homogeneous situations with identical distances $s_i = s$ and velocities $v_i = V$:

$$Q_f(\rho) = \rho V(1/\rho). \quad (6)$$

However, one can show [6] that this homogeneous traffic flow is *linearly unstable*, if

$$\frac{dV(s)}{ds} > \frac{1}{2\tau}. \quad (7)$$

As a consequence, arbitrarily small perturbations, inhomogeneities, or fluctuations of the traffic flow will be amplified, until a congested traffic state with areas of high density and low velocity has evolved.

The *linearly stable* regimes, in which the above condition is not fulfilled and traffic is insensitive to small disturbances, are themselves divided into

- an absolutely stable regime, in which even large perturbations fade away,
- a metastable regime adjacent to the linearly unstable regime [7].

The metastable regime is characterized by the decay of those perturbations, which are smaller than a certain density-dependent “critical amplitude”, while larger perturbations are growing (“nucleation effect”) [7,8].

Stable traffic flow is only found as long as the function $V(s)$ changes little with the distance s , i.e., when then the vehicle density is low (“free” traffic) or high (since the velocity is anyway small, then). However, for traffic densities around 25 vehicles per kilometer and lane (corresponding to velocities below about 85 km/h), the “optimal velocity” is rapidly falling with decreasing distance. Therefore, small changes are eventually amplified, until a traffic jam is formed, even if there is no accident or bottleneck. One could say that such a “phantom traffic jam” is the consequence of a chain reaction based on individual over-reactions: The finite adaptation time τ causes a delayed reaction to changes of the traffic situation, so that a vehicle gets too close to its leading vehicle, which requires an over-braking. If the following vehicles already arrive before the correspondingly increased disturbance can be compensated for, the successive vehicles will all brake a little bit more. In the end, traffic will reach a standstill, although all vehicles would like to drive fast.

If the inflow is not significantly reduced, the traffic jam will not dissolve, because accelerating vehicles lose time when they leave the downstream front of congested traffic. The correspondingly increased time gaps imply a considerably reduced outflow from congested traffic, compared to the maximum flow $Q_{\max}$ of homogeneous traffic defining the *theoretical* capacity of the road. As a consequence, the outflow Q_{out} from congested traffic is a second, *dynamical* capacity, which is something like a self-organized “natural” constant of traffic flow [9].

3.2 The Rich Variety of Congested Traffic States

Most cases of traffic congestion on an n -lane freeway are observed upstream of on-ramps or other bottlenecks. They can be triggered by perturbations

significantly below the theoretical capacity, as soon as the sum of the upstream flow Q_{up} and the inflow $\Delta Q = Q_{\text{in}}/n$ per lane exceeds the dynamical capacity Q_{out} : If a disturbance leads to temporary congestion, the drivers must accelerate again and suffer some time delay, which reduces the capacity to Q_{out} . Therefore, the following vehicles will queue up, and the temporary perturbation grows to form a persistent kind of congestion. The initial perturbation can even be a temporary *reduction* of the traffic flow and/or vehicle density, which can be caused by temporal variations of the traffic volume or even by vehicles leaving the freeway at some off-ramp: When the traffic density has sufficiently increased to reach the metastable regime, the “negative” perturbation will be amplified if it only exceeds the critical amplitude. While it is small, it will move downstream with the vehicles, so one could hope it would pass the bottleneck and leave the system. However, when the density wave grows larger, it will reduce its speed and even change its propagation direction. Once it is fully developed, it moves upstream with constant velocity, since vehicles leave the jam at the downstream front, while new ones join it at the upstream front. Hence, the perturbation returns to the bottleneck like a boomerang and triggers a breakdown of traffic, when it passes the bottleneck in upstream direction, as it thereby reduces the effective capacity to Q_{out} .

If the total traffic volume $Q_{\text{tot}} = (Q_{\text{up}} + \Delta Q)$ is greater than the dynamic capacity Q_{out} , we will automatically end up with a growing vehicle queue upstream of the on-ramp. The traffic flow Q_{cong} resulting in the congested area gives, together with the inflow ΔQ , exactly the outflow Q_{out} , i.e. $Q_{\text{cong}} = (Q_{\text{out}} - \Delta Q)$. One can distinguish the following cases [10] (see Fig. 1): If the density ϱ_{cong} associated with the flow $Q_{\text{cong}} = Q_{\text{f}}(\varrho_{\text{cong}})$ is stable, we find homogeneous congested traffic (HCT) corresponding to typical traffic jams during holiday seasons. For a smaller on-ramp flow ΔQ , the congested flow Q_{cong} is linearly unstable, and we either find oscillating congested traffic (OCT) or triggered stop-and-go traffic (TSG), which often emerges from a spatial sequence of homogeneous and oscillating congested traffic (so-called “pinch effect” [11]). In contrast to OCT, stop-and-go traffic is characterized by a sequence of moving jams, between which traffic flows freely. Each traffic jam triggers the next one by inducing a small perturbation at the ramp, which propagates downstream as long as it is small, but turns back when it has grown large enough (“boomerang effect”). This, however, requires the downstream traffic flow to be linearly unstable. If it is metastable instead (when the traffic volume is further reduced), a traffic jam will usually not trigger a growing perturbation. In that case, one finds either a single moving localized cluster (MLC), or a pinned localized cluster (PLC) at the location of the ramp. The latter requires the traffic flow in the upstream section to be stable, so that no traffic jam can survive there. Finally, for sufficiently small traffic volumes, we find free traffic (FT), as expected.

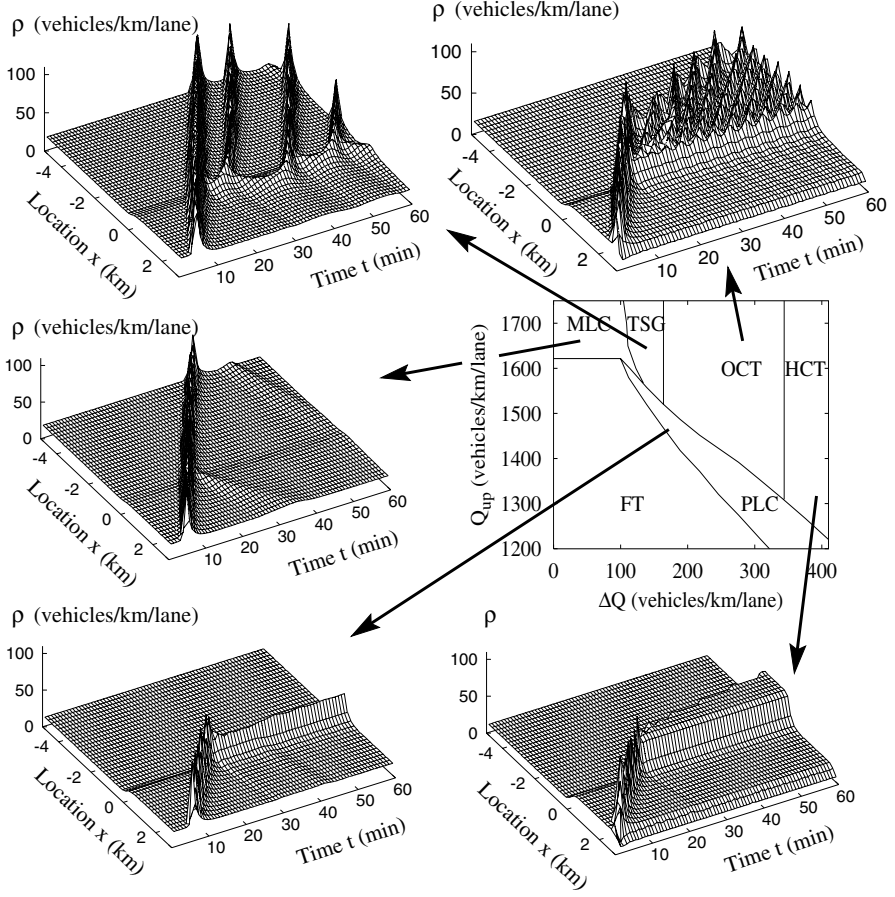


Fig. 1. Representatives of the different congested traffic states, which were induced by a big perturbation travelling upstream, and phase diagram of the traffic states as a function of the (upstream) traffic volume Q_{up} on the freeway and the strength ΔQ of a bottleneck at location $x = 0$ km, e.g. an on-ramp with inflow ΔQ per freeway lane (center right)

4 Collective Pedestrian Dynamics

Further interesting phenomena are found in two-dimensional pedestrian traffic. As the acceleration time $\tau_i \approx 0.5$ s of pedestrians is very short, the instability phenomena observed in vehicular traffic do not appear.

We describe the *psychological tendency* of two pedestrians i and j to stay away from each other by a repulsive interaction force $A_i \exp[(r_{ij} - d_{ij})/B_i] \mathbf{n}_{ij}$, where A_i and B_i are constants. $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ denotes the distance between the pedestrians' centers of mass, and $\mathbf{n}_{ij} = (n_{ij}^1, n_{ij}^2) =$

$(\mathbf{x}_i - \mathbf{x}_j)/d_{ij}$ is the normalized vector pointing from pedestrian j to i . If their distance d_{ij} is smaller than the sum $r_{ij} = (r_i + r_j)$ of their radii r_i and r_j , the pedestrians touch each other. In this case, we assume two additional forces inspired by granular interactions [12,13], which are essential for understanding the particular effects in panicking crowds: a “body force” $k(r_{ij} - d_{ij}) \mathbf{n}_{ij}$ counteracting body compression and a “sliding friction force” $\kappa(r_{ij} - d_{ij}) \Delta v_{ji}^t \mathbf{t}_{ij}$ impeding *relative* tangential motion, if pedestrian i comes close to j . Herein, $\mathbf{t}_{ij} = (-n_{ij}^2, n_{ij}^1)$ means the tangential direction and $\Delta v_{ji}^t = (\mathbf{v}_j - \mathbf{v}_i) \cdot \mathbf{t}_{ij}$ the tangential velocity difference, while k and κ represent large constants. In summary, we have

$$\begin{aligned} \mathbf{f}_{ij} = & \{A_i \exp[(r_{ij} - d_{ij})/B_i] + k\Theta(r_{ij} - d_{ij})\} \mathbf{n}_{ij} \\ & + \kappa\Theta(r_{ij} - d_{ij}) \Delta v_{ji}^t \mathbf{t}_{ij}, \end{aligned} \quad (8)$$

where the function $\Theta(x)$ is zero, if the pedestrians do not touch each other ($d_{ij} > r_{ij}$), otherwise equal to the argument x .

The interaction with the boundaries is treated analogously, i.e., if d_{iW} means the distance to wall W , $\mathbf{n}_{iW}$ denotes the direction perpendicular to it, and $\mathbf{t}_{iW}$ the direction tangential to it, the corresponding interaction force with the wall reads

$$\begin{aligned} \mathbf{f}_{iW} = & \{A_i \exp[(r_i - d_{iW})/B_i] + k\Theta(r_i - d_{iW})\} \mathbf{n}_{iW} \\ & - \kappa\Theta(r_i - d_{iW}) (\mathbf{v}_i \cdot \mathbf{t}_{iW}) \mathbf{t}_{iW}. \end{aligned} \quad (9)$$

4.1 Lane Formation and “Freezing by Heating”

For normal, relaxed situations with small fluctuation amplitudes η_i , our microsimulations of counterflows in corridors reproduce the empirically observed *formation of lanes* consisting of pedestrians with the same desired walking direction, see Fig. 2a [14,15,4]. If we do not assume periodic boundary conditions, these lanes are dynamically varying (see the Java applet at <http://www.helbing.org/Pedestrians/Corridor.html>). Their number depends on the width of the street [14], on pedestrian density, and on the noise level. Interestingly, one finds a *noise-induced ordering* [16]: Compared to small noise amplitudes, medium ones result in a more pronounced segregation (i.e., a smaller number of lanes), while large noise amplitudes lead to a “freezing by heating” effect (see Fig. 2b).

The above, symmetric model can explain lane formation without assuming a preference for *any* side [15,17]. The most relevant point is the higher relative velocity of pedestrians walking in opposite directions. As a consequence, they have more frequent interactions until they have segregated into separate lanes. The resulting collective pattern of motion minimizes the frequency and strength of avoidance maneuvers, if fluctuations are weak. This state is related with a maximum efficiency of motion. One can find such *optimal self-organization* for a large class of driven many-particle systems.

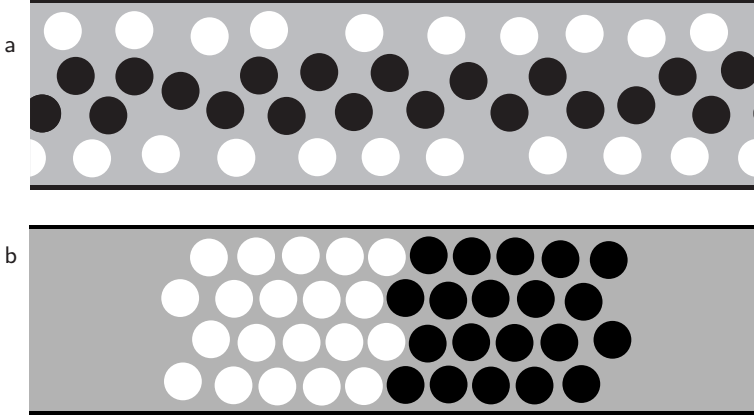


Fig. 2. (a) Formation of lanes in initially disordered pedestrian crowds with opposite walking directions and small noise amplitudes η_i (after [17,18,4]; cf. also [14,15]). White disks represent pedestrians moving from left to right, black ones move the other way round. (b) For sufficiently high densities and large fluctuations, we observe the noise-induced formation of a crystallized, “frozen” state (after [17,18,4])

That is, if these systems self-organize at all, they simultaneously tend to globally optimize their state [15].

To reflect the effect of getting nervous in panic situations, we will now assume that the individual level of fluctuations is given by

$$\eta_i = (1 - n_i)\eta_0 + n_i\eta_{\max}, \quad (10)$$

where n_i with $0 \leq n_i \leq 1$ measures the nervousness of pedestrian i . The parameter η_0 means the normal and $\eta_{\max}$ the maximum fluctuation strength. It turns out that, at sufficiently high pedestrian densities, lanes are destroyed by an increasing fluctuation strength (which is analogous to the temperature). However, instead of the expected transition from the “fluid” lane state to a disordered, “gaseous” state, a solid state is formed [17]. It is characterized by a blocked situation with a regular (i.e. “crystallized” or “frozen”) structure so that we call this paradoxical transition “*freezing by heating*” (see Fig. 2b). Notably enough, the blocked state has a *higher* degree of order, although the internal energy is *increased* and the resulting state is *metastable* with respect to structural perturbations such as the exchange of oppositely moving particles. Therefore, “freezing by heating” is just opposite to what one would expect for equilibrium systems, and different from fluctuation-driven ordering phenomena in metallic glasses and some granular systems, where fluctuations lead from a disordered *metastable* to an ordered *stable* state. The precondition for the unusual freezing-by-heating transition are the additional driving term $v_i^0 \mathbf{e}_i^0 / \tau_i$ and the dissipative friction $-\mathbf{v}_i / \tau_i$, while the sliding friction force is not required.

4.2 Collective Phenomena at Bottlenecks

In simulations of bottlenecks like doors, oscillatory changes of the passing direction are observed, if people do not panic [19,18,4] (see Fig. 3). Once a pedestrian is able to pass the narrowing, pedestrians with the same walking direction can easily follow. Hence, the number and “pressure” of waiting and pushing pedestrians becomes less than on the other side of the narrowing where, consequently, the chance to occupy the passage grows. This leads to a deadlock situation which is followed by a change in the passing direction. Now, we will focus on *uni-directional* flows at bottlenecks. The simulated outflow from a room turns out to be well-coordinated and regular, if the desired velocities $v_i^0 = v_0$ are normal. However, for desired velocities above 1.5 m/s, i.e. for people in a rush, we find an irregular succession of arch-like blockings of the exit and avalanche-like bunches of leaving pedestrians, when the arches break (see Fig. 4a,b). This phenomenon agrees with empirical observations and is comparable to intermittent clogging found in granular flows through funnels or hoppers [12,13].

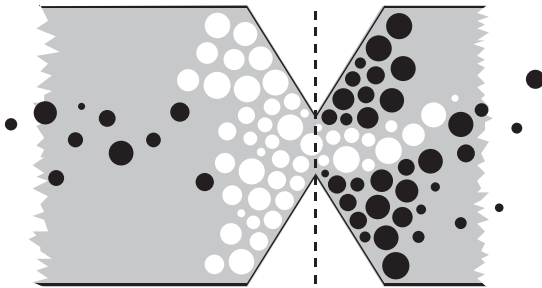


Fig. 3. Oscillations of the passing direction at a bottleneck (after [4,18]; cf. also [14]). Dynamic simulations are available at <http://www.helbing.org/Pedestrians/Door.html>

4.3 “Faster-Is-Slower Effect” Due to Impatience

Since clogging is connected with delays, trying to move faster (i.e., increasing v_i^0) can cause a smaller average speed of leaving, if the friction parameter κ is large enough. This “faster-is-slower effect” is particularly tragic in the presence of fires, where fleeing people reduce their own chances of survival.

Since our friction term has, on average, no deceleration effect in the crowd if the walls are sufficiently remote, the arching underlying the clogging effect requires a *combination* of several effects:

1. slowing down due to a bottleneck such as a door and
2. strong inter-personal friction, which becomes dominant when pedestrians get too close to each other.

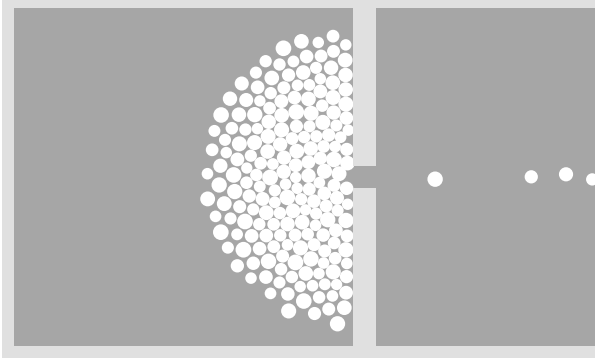


Fig. 4. Panicking pedestrians come so close to each other, that their physical contacts lead to the build up of pressure and obstructing friction effects. This results in temporary arching and clogging related with inefficient and irregular outflows [19,4,18]

Consequently, the danger of clogging can be minimized by avoiding bottlenecks in the construction of stadia and public buildings. Notice, however, that jamming can also occur at widenings of escape routes! This surprising result originates from disturbances due to pedestrians, who expand in the wide area because of their repulsive interactions or try to overtake each other. These squeeze into the main stream again at the end of the widening, which acts like a bottleneck and leads to jamming.

Significantly improved outflows can be reached by columns placed asymmetrically in front of the exits, which also prevent the build up of fatal pressures (see the Java applets at <http://angel.elte.hu/~panic/>).

4.4 “Phantom Panics”

Sometimes, panics have occurred *without* any comprehensible reasons such as a fire or another threatening event (e.g., in Moscow, 1982; Innsbruck, 1999). Due to the “faster-is-slower effect”, panics can be triggered by small pedestrian counterflows [19,4,18], which cause delays to the crowd intending to leave. Consequently, stopped pedestrians in the back, who do not see the reason for the temporary slowdown, are getting impatient and pushy. In accordance with observations [20], one may describe this by increasing the desired velocity, for example, according to the formula

$$v_i^0(t) = [1 - n_i(t)]v_i^0(0) + n_i(t)v_i^{\max}. \quad (11)$$

Herein, $v_i^{\max}$ is the maximum desired velocity and $v_i^0(0)$ the initial one, corresponding to the expected velocity of leaving. The time-dependent parameter

$$n_i(t) = 1 - \frac{\bar{v}_i(t)}{v_i^0(0)} \quad (12)$$

reflects the nervousness, where $\bar{v}_i(t)$ denotes the average speed into the desired direction of motion. Altogether, long waiting times increase the desired velocity, which can produce inefficient outflow. This further increases the waiting times, and so on. This tragic feedback can eventually trigger so high pressures that people are crushed or falling and trampled.

5 Summary and Outlook

One can say that physics presently contributes significantly to a deeper understanding of problems from traffic dynamics, biology, and the socio-economic sciences. It has been found that the general methods from statistical physics and non-linear dynamics are most useful in these areas. In particular, many new insights into the often surprising phenomena in living systems have been gained thanks to the relatively young paradigm of active many-particle systems, which was presented in a general mathematical framework, here.

It would be desirable to develop a statistical mechanics of self-driven many-particle systems on this basis, as it already exists for momentum- and energy-conserving systems. Are there also general principles such as entropy maximization or relations like the partition function, from which the properties of the respectively resulting macroscopic states can be derived? Up to now, this has been answered only for specific non-equilibrium systems. However, this field is not only of interest for fundamental research. Thanks to various scientific projects with industrial partners, the insights from physical traffic theory are meanwhile adopted in traffic optimization strategies. Examples are adaptive speed limits, intelligent on-ramp controls, incident detection methods, traffic prognoses, and driver assistance systems. Active many-particle systems are, therefore, a good example for a field, in which applied and fundamental research are close to each other. Moreover, many results have quite general implications, in particular for biological and socio-economic systems. They are partly transferable to individuals who compete for limited resources like time, money, energy, etc. [16,18].

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